Morphology from node-level measurements. Can a cell-level phenotype (CalMorph) be predicted from the genotype, and does a measured gene-level modality (expression, proteome) help when revealed as input? Transductive: every validation strain carries the revealed modality.

a ✓

The strain sets
three panels and their overlaps: morphology 4,695, expression 1,554, proteome 4,476; morph. and expr. 1,439, morph. and prot. 4,281, all three 1,321. Counts, no statistic.

results/morph_v24_dataset.json

b ✓

The target: 116 moving features
per-feature replicate reliability (122 wild-type profiles against 4,718 mutants) as a histogram; the 0.5 threshold; ceiling = sqrt(reliability), mean 0.847 on the moving set, 0.611 on all 278.

results/morphology_feature_ceiling.csv, calmorph_moving_features.json

c ✓

Genotype only
bars: ceiling; sequence ridge and neighbor average; the CGT off the perturbed CLS (twelve split seeds, window mean, sd). Pearson per feature across held-out strains, moving features.

results/morph_v24_readout.json (M_cls, M_pool)

d ■

Measured modality revealed
paired by split seed: the strain's own proteome (expression) revealed against another strain's, and against genotype only on the same strains; twelve seeds; beside the ridge from the same measurement.

morph_v24_readout.json (CM_prot, CM_protperm, MP_cls; CMP_expr, CMP_exprperm); in flight

e ✗

Per feature
held-out Pearson of the best revealed arm against each feature's reliability; which features a measured modality reaches and which stay at the floor. Dots, 116 features, one seed by rule.

a per-feature dump of the CM_prot checkpoint; not written yet

f ✓

The conditioning triangle
the same contrast on the other two sides: proteome given expression +0.054 (11 of 11), expression given proteome +0.033 (10 of 11), both below their ridge; morphology joins as the third vertex.

results/joint_checkpoint_readout.json (v20), the covariation files

g ✗

Natural variation: the 1,011 isolates
the same conditioning on wild strains: Caudal 2024 expression (built, 943 isolates) and Muenzner 2024 proteome (next build, 796 isolates); morphology not measured on the panel, so the target there is the proteome from expression and the reverse. **Planned; likely not in time for this round.**

the expansion document, next-build band; no result file

✓ ready: can be drawn now from committed results ■ partial, or needs an author decision ✗ not done, placeholder